

Automatic Image Caption Generator

A COURSE END PROJECT REPORT ON DEEP LEARNING

(COURSE CODE –A8709)

Submitted in the Partial Fulfilment of the
Requirements for the Award of the Degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (AI&ML)

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VARDHAMAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

Affiliated to **JNTUH**, approved by **AICTE**, accredited by **NAAC** with **A++ Grade**, **ISO 9001:2015**

Certified, Kacharam, Shamshabad, Hyderabad – 501218, Telangana, India

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI&ML)

CERTIFICATE

This is to certify that the Course End Project titled **Automatic image caption generator** is carried out by H. Uday Kumar-24885A6609, K. Pranay-24885A6610, M. Madhan-24885A6611 towards **Deep Learning** course in partial fulfillment of the requirements for the award of degree of Bachelor of Technology in Computer Science and Engineering (AI&ML) during the Academic Year 2025-2026

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DECLARATION

We hereby declare that the project titled **Automatic image caption generator**, submitted to Vardhaman College of Engineering (Autonomous), affiliated with Jawaharlal Nehru Technological University Hyderabad (JNTUH), in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering (AI & ML), is the result of original work carried out by us. We further certify that this project report, either in full or in part, has not been previously submitted to any university or institute for the award of any degree or diploma.

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Abstract

The exponential growth of digital imagery on the internet has created a pressing need for systems that can understand and describe visual content. Automatic Image Captioning, a key task in the intersection of Computer Vision and Natural Language Processing (NLP), aims to generate descriptive and contextually accurate textual captions for images. This project presents the design and implementation of a user-friendly web application for automatic image captioning. The application leverages the BLIP (Bootstrapping Language-Image Pre-training) model developed by Salesforce, through the Hugging Face transformers library. The web interface is built using Streamlit, a rapid application development framework for data science and machine learning. The system allows users to upload images in common formats (JPG, JPEG, PNG) and receive AI-generated descriptive captions in real-time. Key features of the application include model caching for performance optimization, dynamic support for both CPU and GPU computation, and a responsive, visually appealing user interface. This report documents the entire project lifecycle, including a literature review of relevant models, a detailed explanation of the methodology and system architecture, a discussion of the results, and conclusions with suggestions for future enhancements. The developed application successfully demonstrates the practical deployment of a sophisticated deep-learning model for an accessible and useful AI-powered task. The application focuses exclusively on the BLIP base model for caption generation.

Keywords - Image Captioning, BLIP, Streamlit, Hugging Face, Deep Learning, Computer Vision, Natural Language Processing, Web Application.

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Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
BLIP	Bootstrapping Language-Image Pre-training
CPU	Central Processing Unit
GPU	Graphics Processing Unit
NLP	Natural Language Processing
UI	User Interface
API	Application Programming Interface
PIL	Python Imaging Library
SOTA	State-of-the-Art